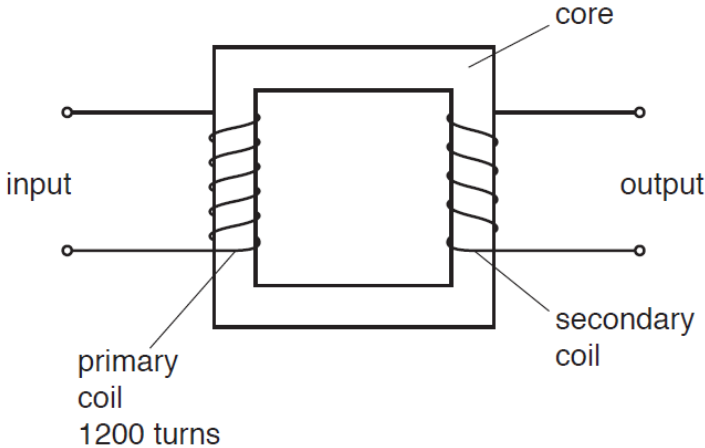


6	An ideal transformer is shown in Fig. 6.1.  Fig. 6.1	
	(a) Explain why an electromotive force (e.m.f.) is not induced at the output when a constant direct voltage is at the input. 	[2]
	(a) Solution: There is induced e.m.f. <u>only when the magnetic flux in the core or coil is changing.</u> However, the <u>constant voltage provides a constant magnetic flux.</u>	1 1
	Examiner's Comments: This question was fairly well done as students generally showed understanding of the link between constant voltage, constant magnetic flux (linkage) and no e.m.f. induced. Those who did not achieve full credit or credit at all either thought that the flux linkage will be zero, or failed to link "constant magnetic field" to constant flux linkage.	

- (b)** An alternating voltage of peak value 150 V is applied across the 1200 turns of the primary coil. The variation with time t of the e.m.f. E induced across the secondary coil is shown in Fig. 6.2.

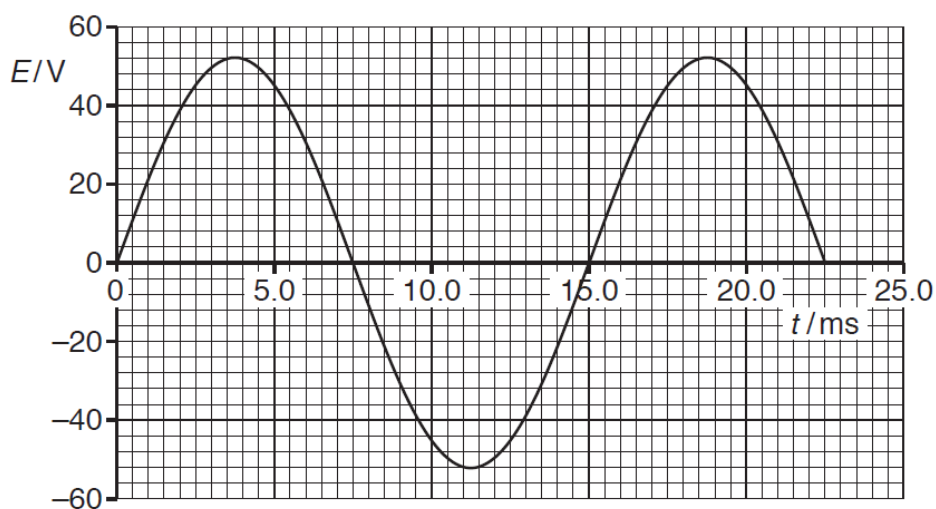


Fig. 6.2

Use data from Fig. 6.2 to

- (i)** calculate the number of turns of the secondary coil,

number of turns =

[2]

- (i) Solution:**

$$\begin{aligned}\frac{N_s}{N_p} &= \frac{V_s}{V_p} \\ N_s &= \frac{V_s}{V_p} N_p \\ &= \frac{50}{150} (1200) \\ &= 400 \text{ turns}\end{aligned}$$

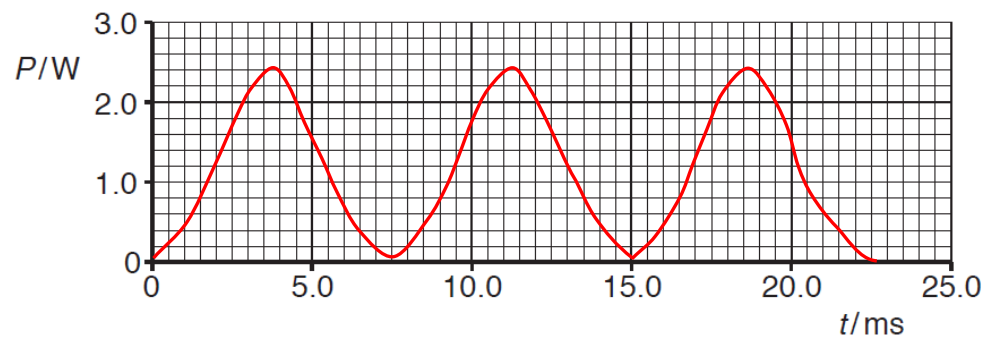
1

1

- (ii)** state one time when the magnetic flux linking the secondary coil is a maximum.

time = ms

[1]



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Section B

Answer **one** question from this Section in the spaces provided.

7	(c)	State the first law of thermodynamics	
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